

Course Syllabus ▲▼

ENAE 694

Spacecraft Communications

Fall 2026 | Syllabus | August 31, 2026

Instructor Dave Israel

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Meets Monday evenings

Time and room 5:00-8:00 p.m.; Martin
Hall 1110

First class August 31, 2026

Final assessment Final project due on
the university-assigned final-exam date

Course purpose

Spacecraft communications moves commands, telemetry, science, navigation measurements, and other information across imperfect connections. I do not expect everyone to become a spacecraft communications systems engineer. My goal is that you leave with an understanding and appreciation of their work: the concepts, factors, and trades behind their decisions and their effects on the mission.

Governing idea Everything is a trade. Equations support engineering judgment; they do not replace it.

Course arc

1	2	3	4	5
Mission needs	Individual links	Coverage	Data delivery	Network architecture

Learning outcomes

- Translate mission operations, data volume, latency, completeness, coverage, and resilience needs into measurable communications design targets and constraints.

- Construct and interpret link budgets using power, path loss, antenna gain, noise, modulation, coding, and implementation margin.
- Explain major antenna, pointing, spectrum, coding, multiple-access, relay, RF-versus-optical, and ground-system trades.
- Evaluate coverage and availability using supplied contact data, including total availability, longest outage, and operational consequences.
- Analyze buffering, contact schedules, store-and-forward delivery, latency, and routing in intermittently connected networks.
- Recommend and defend an end-to-end spacecraft communications architecture that closes its representative links and meets mission-level service needs.

How we will work together

Class meetings will combine short explanations with worked engineering examples. We will calculate representative cases, change assumptions, predict what should happen, and talk about what each result means. Homework will build on work we have already done together rather than introducing an unfamiliar calculation cold. Students in section AED1 participate asynchronously; joining the live class virtually is optional, and no graded work requires real-time attendance.

For coverage work, I will supply the orbit/contact data and guided analysis tools. The important part is understanding what the results mean for the mission, not building an orbit propagator. Some homework may also give you useful pieces to carry into the final project.

How your work will be assessed

Category	What it includes	Purpose
Homework and applied work - 70%	Calculations, interpretation, and engineering judgment based on material introduced in class	Practice, feedback, and revision until the work is correct
Final project - 30%	Communications design and presentation materials for a mission concept review	Synthesis, engineering judgment, and communication

Grading model

Homework and related applied work account for 70% of the course grade, and the final project accounts for 30%. There is no separate attendance or engagement grade and no timed midterm or final examination. I know that many students are balancing this course with jobs and other

responsibilities, so I will be flexible where I reasonably can. I will provide feedback on homework, and you may revise and resubmit it until it is correct. The goal is for everyone to learn the material, not for an early mistake to become permanent.

Final project: the mission concept review

The final project will put you in the role of the communications systems engineer for a proposed mission. You will develop a preliminary communications design and prepare the presentation materials you would bring to a mission concept review. The point is to show how concepts, constraints, and trades lead to a defensible design, not to produce a heroic amount of documentation. Detailed expectations will be provided with the assignment. The submission will include a slide deck and separate written material. It will be due on the university-assigned final-exam date, in lieu of a final exam. There is no live presentation or recorded narration for either section.

Class schedule

The schedule below reflects UMD's Fall 2026 academic calendar and the Jewish-holiday review. Topics and assignment dates may shift modestly as the lecture materials are rebuilt.

Date	Class	Topic	Applied focus
Aug. 31	1	The Spacecraft Communications Problem	Mission needs, constraints, data volume, latency, completeness, and the end-to-end problem.
Sept. 7	-	No class - Labor Day	University holiday.
Sept. 14	2	Received Power and Link Geometry	Decibels, EIRP, free-space loss, received power, and scale from LEO to deep space.
Sept. 21	-	No class - Yom Kippur	No course meeting or deadline.

Date	Class	Topic	Applied focus
Sept. 28	3	Antennas and Pointing	Aperture, gain, beamwidth, pointing loss, mass, coverage, and EIRP trades.
Oct. 5	4	Optical Communications	RF-versus-optical trades; pointing, acquisition and tracking; atmospheric availability; and ground diversity.
Oct. 12	-	No class - Fall Break	University break.
Oct. 19	5	Noise, G/T, and the Complete Link Budget	Noise temperature, G/T, received energy, margin, and a complete link budget.
Oct. 26	6	Signals, Modulation, and Bandwidth	Data rate, bandwidth, power efficiency, implementation complexity, and waveform trades.
Nov. 2	7	Demodulation, BER, and Coding	Coding overhead and gain; trades among EIRP, rate, margin, and reliability.
Nov. 9	8	Multiple Access, Crosslinks, and Relays	Shared resources, access, relays, crosslinks, capacity, bottlenecks, and scheduling.
Nov. 16	9	Coverage and Availability	Contact data, availability, outage duration, diversity, and operations.
Nov. 23	10	Spacecraft Data Services and End-to-End Data Flow	Packetization, buffering, service boundaries, and end-to-end data movement.
Nov. 30	11	Routing, Store-and-	Contact schedules, intermittent

Date	Class	Topic	Applied focus
		Forward, and DTN	routing, latency, custody, delivery, and interoperability.
Dec. 7	12	Interoperable Space Networks and Architecture Studio	LunaNet and multi-provider networks; integrate links, coverage, delivery, resilience, complexity, and cost. Final-project studio.

Homework, feedback, and revision

The number, topics, and timing of homework assignments may change as the class develops. I will adjust the work to match our pace and to give us more practice where it is useful. Assignments and deadlines will be posted in ELMS with reasonable notice, and homework deadlines will not depend on live attendance. After I return feedback, you may revise and resubmit your work until it is correct.

What I will look for in your work

Show the reasoning. A numerical answer without assumptions, units, or interpretation is incomplete.

Use judgment. When a choice is required, identify the trade and explain why the selection fits the mission.

Check plausibility. Use dimensional analysis, order-of-magnitude checks, and comparison cases.

Document tools. Spreadsheets, scripts, and AI-assisted work must be understandable, reproducible, and consistent with the assignment's allowed-collaboration rules.

Design end to end. A link that closes is necessary, but it is not sufficient if coverage, data flow, latency, operations, or interoperability fail.

Course materials and tools

Lecture slides, worked examples, homework templates, reference data, and analysis tools will be distributed through ELMS. There are no required or recommended readings at the start of the course, although I may share occasional readings during the semester. Office hours will be announced in ELMS. Students should expect to use a spreadsheet and, when supplied, a guided notebook or similar analysis tool. Programming is not itself a course objective.

Collaboration and AI-assisted work

Collaboration rules will be stated on each assignment. Discussion of concepts and approaches may be encouraged, while submitted calculations, explanations, and design choices must represent the student's own understanding. Any use of generative AI must follow the assignment-specific instructions and must not replace the student's ability to explain or defend the work.

Attendance, absences, and religious observance

For the campus section, being in class will make it easier to work through examples and ask questions, but attendance is not a separate part of the grade. Section AED1 is asynchronous: students may join the live class virtually, but attendance in person or in real time is never required for lectures, assessments, or class activities. Recordings and any activity prompts will be posted through ELMS, and short class activities will remain available to AED1 students for at least 48 hours after the recording is available. Please let me know promptly about excused absences or religious observances so we can make a reasonable plan without an academic penalty.